



# Brendon Mutema

Electrical & Computer Engineer | Python, 5G/Wi-Fi Coexistence & Applied ML | MScEng Candidate, UCT

University of Cape Town

City of Cape Town, Western Cape, South Africa · Contact info

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**Open to work · Recruiters only**

Canada +4 more | On-site · Hybrid · Remote

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## About

As a recent Electrical and Computer Engineering graduate and an incoming Master's student at the University of Cape Town, I am passionate about solving complex challenges at the intersection of wireless communications and system engineering. My academic journey has provided me with a robust foundation in software development, system analysis, and practical hardware implementation.

My final year thesis on 5G/Wi-Fi coexistence allowed me to develop a discrete-event simulator from scratch in Python, honing my skills in system modeling, performance optimization, and data-driven analysis. This experience, combined with my upcoming Master's research on 5G/satellite coexistence for ICASA, demonstrates my commitment to tackling real-world spectrum management problems.

I am a proactive and resilient learner always eager to master new technologies. I thrive in collaborative environments and am now seeking graduate or internship opportunities where I can contribute my technical skills and grow as an engineer. Let's connect!

**Top skills**
Python (Programming Language) • Wireless Communications • Machine Learning • 5G NR and Wi-Fi Coexistence →

## Activity

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**Brendon Mutema** • You
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Full technical stack complete: **20/20 certifications cleared**. 🚀 ... more

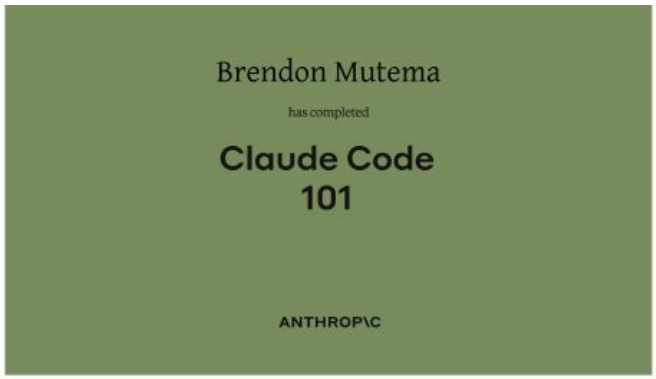





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New Milestone: 10 technical certifications complet 🚀 ... more






Moutloatsi Setlogelo and 3 others

101 impressions View analytics




134 impressions View analy

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## Experience

### Machine Learning Internship
FlyRank AI · Internship
Jun 2026 - Sep 2026 · 4 mos
Pretoria, Gauteng, South Africa · Remote

Selected for the Machine Learning track within a global cohort of 20,000+ applicants. My focus is on mastering the practical implementation and deployment of AI models within industrial environments. I am applying the systems-thinking and Python simulation skills from my BSc/MSc at UCT to build production-ready ML capstones.



**FlyRank - The Autopilot for Organic Growth**

**Python (Programming Language)**

### Tutor
MindShift Learning · Part-time
Mar 2026 - Aug 2026 · 6 mos
City of Cape Town, Western Cape, South Africa · Remote

Leveraging my background in Electrical and Computer Engineering to provide specialized academic support in STEM subjects, including Mathematics and Engineering principles.

Key responsibilities include:
• Translating complex technical concepts into accessible, structured learning materials.
• Mentoring students in advanced problem-solving methodologies and analytical thinking.

• Developing technical communication skills essential for bridging the gap between high-level theory and practical application.

**Technical Communication, Analytical Skills and +2 skills**

### Engineering Intern

Mamoshe Investment Group (Pty) Ltd · Internship

Aug 2025 - Sep 2025 · 2 mos

Polokwane, Limpopo, South Africa · On-site

Completed six weeks of practical training (EEE3000X) contributing to the CCTV installation project at Waterberg TVET College. The role involved hands-on implementation and technical support, including installing, terminating, and testing CAT6 network cables according to the T568B standard to ensure reliable connectivity. Assisted senior engineers with configuring network switches and implementing VLANs to isolate and secure CCTV traffic from the main data network. Applied IEEE 802.3at (PoE+) principles to verify power delivery to IP cameras, considering switch power budgets and cable length limitations. Also supported prototype testing and provided general IT system assistance as required by the project team.





**Technical Support, Problem Solving and +2 skills**

### Vacation Work Trainee

Cape Peninsula University of Technology · Internship

Dec 2021 · 1 mo

City of Cape Town, Western Cape, South Africa · On-site

Completed a two-week intensive training program at Cape Peninsula University of Technology, covering Welding, Pneumatics, Metrology, Fitting and Turning, Robotics, and Basic Engineering Hand Skills. During the vacation work training program, I applied theoretical knowledge to practical tasks and developed hands-on technical skills that strengthened my understanding of industrial processes and engineering workshop practices.




**Brendon E T Mutema - Certificate of Training.pdf**
Official Certificate of Training awarded to Brendon E. T. Mutema by Cape Peninsula University of Technology, confirming successful completion of the ...

**Fitting and Turning, Robotics and +4 skills**

## Education

### University of Cape Town

Master of Science - MS, Electrical Engineering

Feb 2026 – Present

### University of Cape Town

Bachelor of Science - BSc, Electrical & Computer Engineering

Jan 2019 – Jun 2025

Activities and societies: Member of the UCT IEEE (Institute of Electrical and Electronics Engineers) Student Branch

Completed a comprehensive NQF Level 8 degree with a curriculum focused on software development, embedded systems, electronics, and communication systems. The program emphasized project-based learning and a systematic approach to engineering problem-solving.

Final Year Thesis: Coexistence Strategy for 5G New Radio and Wi-Fi
My capstone project involved designing and validating an "Optimized Gap Mode" strategy to mitigate interference in shared spectrum. Key achievements included:
• Developed a discrete-event simulator from scratch using Python (SimPy) to model complex network interactions.
• Achieved near-perfect fairness (Jain's Fairness Index > 0.97) and balanced channel occupancy in simulations, demonstrating a significant improvement over baseline models.

**Project Management, Version Control Tools and +18 skills**





Show all 5 educations →

## Licenses & certifications (20)

### Claude with Google Cloud's Vertex AI
Anthropic
Issued Aug 2026

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**Vertex AI**




**Claude on Google Cloud - Certificate of Completion**
Technical certification (93 lessons) on integrating and scaling AI models within the Google Cloud ecosystem. Expertise in Vertex AI configuration, systematic ...

### Claude with Amazon Bedrock
Anthropic
Issued Aug 2026

Show credential

**Amazon Web Services (AWS), Amazon Bedrock and +2 skills**




**Claude with Amazon Bedrock - Certificate of Completion**
Technical certification (83 lessons) on architecting and deploying AI solutions within the AWS ecosystem. Validated expertise in AWS Bedrock API integration...

Show all 20 licenses →

## Projects

### Final Year Thesis: Coexistence Strategy for 5G New Radio and Wi-Fi

Jan 2025 – Jun 2025

My capstone project for my Bachelor of Science in Engineering. I designed and validated an "Optimized Gap Mode" strategy to mitigate interference and ensure fair coexistence between 5G NR-U and Wi-Fi networks in shared spectrum.
To achieve this, I developed a discrete-event simulator from scratch using Python and the SimPy library to model complex network interactions and analyze performance metrics. The final strategy successfully achieved near-perfect fairness (Jain's Fairness Index > 0.97) and balanced channel occupancy (~45-50% per technology) in simulations, demonstrating a significant improvement over baseline models.

**Python (Programming Language), Wireless Communications...**

Associated with University of Cape Town

## Skills (100)

You've added the maximum number of skills

**5G NR and Wi-Fi Coexistence**

**Machine Learning**

Show all →

## Interests

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## Causes

Education • Environment • Science and Technology • Poverty Alleviation • Social Services

## Profile language

English

## Public profile & URL

www.linkedin.com/in/brendon-mutema

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Megchem

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### Someone at Harare Institute of Technology

Harare Institute of Technology

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**Government Policy Specialist in the Professional Organizations industry**

Rhevia

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MSc Student

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Student at University of Cape Town

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**Manare Rammutla** · 2nd
MSc (Eng) Candidate, Telecommunications | Electrical and Computer Engineering Graduate | University of Cape Town

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**Neo Shibambo** · 3rd+
Student at University of Cape Town

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**Sampath Jayalath** · 2nd
Senior Lecturer, University of Cape Town | IEEE CertifAIEd™ Authorized Assessor | Accidental Musician

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